

Reproducibility I

Reproducibility is not a claim this manuscript makes about itself; it is a property the code enforces on every run. Every number that appears below the Determinism heading — `f84a8f9dbcb18e37`, `42`, `493`, `96.28` — is a token substituted at render time by `src/manuscript_variables.py::generate_variables()` from a live grammar load and a live pytest run (`src/manuscript_variables.py::measure_test_summary()`). Neither function accepts a hardcoded literal as a fallback: if the subprocess pytest run cannot be parsed, `measure_test_summary` returns the string "pending" for both the test count and the coverage percentage rather than a plausible-looking number [`@reproducible_builds`]. This design choice is a direct consequence of an earlier failure mode in this project — a manuscript draft that stated fixed values ("Tests: 371 · Coverage: 99.94%") in prose instead of through the token pipeline. A number that cannot be traced back to a specific function call in `src/` is, for the purposes of this project, not a number this manuscript is permitted to assert.

Reproducibility II

Determinism guarantee

Given the same `manuscript/config.yaml` grammar definition (grammar hash `f84a8f9dbcb18e37`), the same seed (42), and the same `template_autopoiesis` source tree, `scripts/autopoiesis.py expand` produces byte-identical selections on every invocation, and `materialize` produces a byte-identical child project tree. The determinism chain has three concrete steps, each implemented as a pure function with no random or wall-clock input:

1. **Selection.** For every grammar slot, `src/expand.py::_digest_index()` builds the key `f"{seed}\x1f{slot_name}\x1f{ordinal}\x1f{'.'.join(options)}` (a unit separator, `\x1f`, joins the fields so that no combination of seed/name/ ordinal/option values can collide across a field boundary), hashes it with SHA-256, and reduces the first eight bytes of the digest modulo `len(options)` to obtain the chosen option's index. Because the digest is a pure function of (seed, slot_name, ordinal, options), the same grammar and seed walk every slot to the same choice on